

Climate-driven dengue early warning in Indonesia: an East Java case study with a seven-province comparison

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2025

Abstract

Dengue transmission in Indonesia is strongly climate-sensitive, yet routine surveillance remains largely reactive. As a compact, reproducible demonstration of the modelling that underpins climate-based early-warning systems, I link open monthly dengue surveillance (Open-Dengue v1.3) with open climate reanalysis (Open-Meteo / ERA5) and fit negative-binomial distributed-lag non-linear models (DLNMs) of dengue *incidence*. A single-province analysis for East Java (2004–2024) is extended to a pooled seven-province model spanning Java and Bali, with a population offset and province-specific baselines. Across all provinces the strongest rainfall signal occurs at a delay of two to three months, consistent with the biological lag between rainfall, vector breeding and onward transmission. The pipeline is intended as a portfolio artefact and the foundation for a doctoral programme on operational dengue early warning.

1 Background

Dengue is among the fastest-growing vector-borne diseases worldwide, and Indonesia carries one of the highest burdens in Southeast Asia. Temperature and rainfall govern *Aedes* survival, biting rates and the extrinsic incubation period, with effects that unfold over weeks to months. That delayed structure is precisely what a distributed-lag model quantifies, and it is what makes forecasting with useful lead time possible. This note is a deliberately minimal first step: real surveillance data, an open climate source, and a standard, defensible model, released reproducibly.

2 Data

Monthly, province-level (Admin1) reported dengue case counts were extracted from the Open-Dengue database v1.3, which standardises publicly available surveillance data. Seven provinces with near-complete monthly coverage were retained (East Java, West Java, Central Java, DKI Jakarta, Banten, Bali and DI Yogyakarta), spanning January 2004 to July 2024. Provincial population (BPS Statistics Indonesia census and intercensal projections) was interpolated to monthly

values and used as a model offset, converting counts to incidence. Monthly mean temperature, total rainfall and relative humidity were obtained from the Open-Meteo Historical Weather API (ERA5-based reanalysis) at representative provincial coordinates.¹

3 Methods

Case counts $Y_{p,t}$ in province p and month t were modelled with a negative-binomial GLM,

$$\log \mathbb{E}[Y_{p,t}] = \log N_{p,t} + \alpha_p + \text{cb}_{\text{temp}}(x_{p,t}, \ell) + \text{cb}_{\text{rain}}(r_{p,t}, \ell) + s(\text{month}_t) + f_p(t),$$

where $N_{p,t}$ is population (offset), α_p a province-specific intercept, and each *cross-basis* $\text{cb}(\cdot)$ is a tensor product of a natural-spline exposure–response basis and a natural-spline lag–response basis over lags $\ell = 0, \dots, 6$ months (Gasparrini 2011). Seasonality $s(\cdot)$ uses cyclic terms and the trend $f_p(t)$ a natural spline on time. Negative binomial errors accommodate overdispersion. Estimates are summarised as cumulative relative risks (RR) over the 0–6 month lag window, centred at the median exposure. Two implementations are provided: a self-contained Python version and the canonical `dlnm` R package.

4 Results

4.1 East Java

Dengue incidence is strongly seasonal, peaking in the wet-season months (January–March) and lowest in August–September (Figure 1). The strongest rainfall–dengue cross-correlation occurs at a two-month lag. The cumulative exposure–response indicates rising dengue risk with increasing monthly rainfall across the observed range (Figure 2). The negative-binomial DLNM fits the series closely.

4.2 Seven provinces

Extending to a pooled seven-province incidence model preserves the pattern while revealing between-province heterogeneity. Incidence trajectories differ markedly in level and timing (Figure 3), yet the peak rainfall–dengue association clusters at a two-to-three month lag in every province (Figure 4). This consistency of the delayed rainfall effect, against a backdrop of heterogeneous baseline risk, is exactly the structure an early-warning system can exploit.

4.3 Large-scale drivers: ENSO and IOD

Local weather is itself modulated by large-scale climate modes. Two were added to the East Java model as additional cross-bases: the El Niño–Southern Oscillation, via the Oceanic Niño Index (ONI), and the Indian Ocean Dipole, via the Dipole Mode Index (DMI) — the latter especially relevant to Indonesian rainfall. The estimated exposure–response for ONI (Figure 5) summarises how dengue risk varies across La Niña and El Niño states, cumulated over a six-month lag. Because these indices partly act *through* local rainfall, they are naturally collinear with it; whether they add independent predictive value is an empirical question addressed next.

¹The figures shown here use a clearly-labelled synthetic climate series so the pipeline runs without network access; replacing it with real Open-Meteo data is a single scripted step. Figures generated from synthetic climate are tagged accordingly.

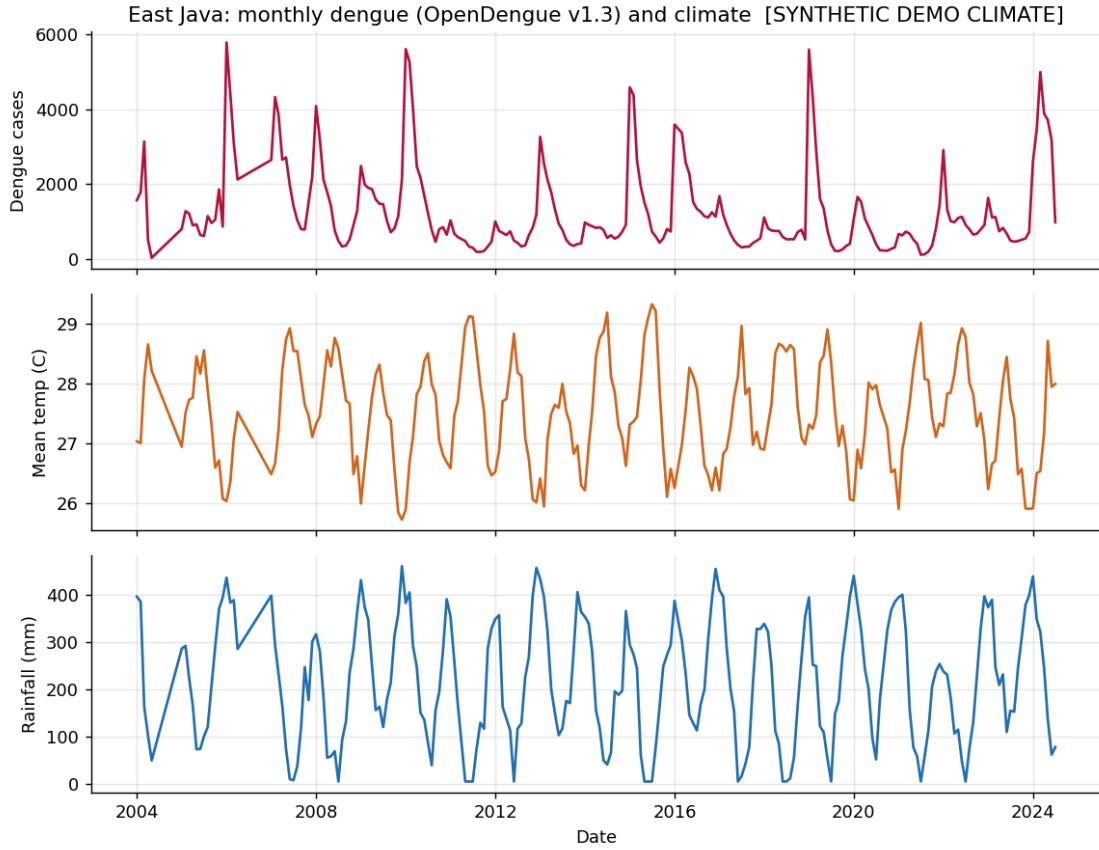


Figure 1: East Java: monthly dengue cases and climate, 2004–2024 (OpenDengue v1.3).

4.4 Forecast skill

To move from association to a genuine early-warning prototype, the model was trained on 2004–2019 and evaluated on the held-out 2020–2024 period, against a seasonal “climatology” baseline containing no climate information. For East Java the climate DLNM improved the out-of-sample forecast substantially (correlation with observed cases 0.70 versus 0.38 for the baseline; an RMSE skill gain of roughly one fifth) and detected a larger share of true outbreak months (Figure 6). Adding ENSO and IOD indices at fixed lags did not improve out-of-sample skill here (correlation 0.65): in this demonstration the indices are largely redundant with local rainfall, and entering them instead as flexible cross-bases *raised* in-sample fit while *lowering* held-out skill — a textbook reminder that fit and forecast are different targets. Skill was also heterogeneous across provinces — unsurprising given single-point climate proxies, the COVID-19 disruption to 2020–2021 surveillance, and the synthetic climate used here. Establishing where and when these drivers add genuine forecast value, with real reanalysis and seasonal-forecast climate, is itself a core research question.

5 Discussion

This analysis reproduces, at small scale, the core building block of a dengue early-warning system: an interpretable, lagged climate–incidence relationship estimated with a standard model, and shown to be reasonably consistent across seven provinces. It is deliberately limited: single climate points per province, no formal forecast evaluation, and surveillance case definitions that vary over the source period. A doctoral programme would address exactly these gaps — fully hierarchical

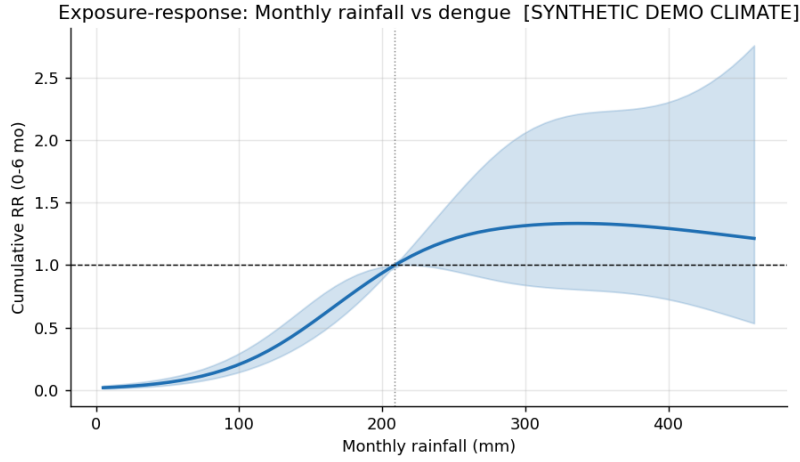


Figure 2: East Java cumulative exposure–response for monthly rainfall (0–6 month lag), relative to the median. Shaded band: 95% CI.

spatio-temporal models, forecast-skill evaluation against held-out epidemics, incorporation of ENSO, urbanisation and mobility, and district-level alert thresholds co-designed with health authorities, and forecast evaluation with proper probabilistic scoring rules and seasonal-forecast climate rather than the single hold-out shown here.

Reproducibility

All code and the real dengue extract are openly available. The analysis runs end-to-end from the command line; real climate is obtained via a single script (Open-Meteo, no API key).

Key references

Gasparrini A. Distributed lag linear and non-linear models in R: the package dlnm. *J Stat Softw.* 2011;43(8). Lowe R, et al. Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil. *Lancet Planet Health.* 2021. Clarke J, et al. A global dataset of publicly available dengue case count data. *Sci Data.* 2024;11:296. Colón-González FJ, et al. Projecting the risk of mosquito-borne diseases in a warmer and more populated world. *Lancet Planet Health.* 2021.

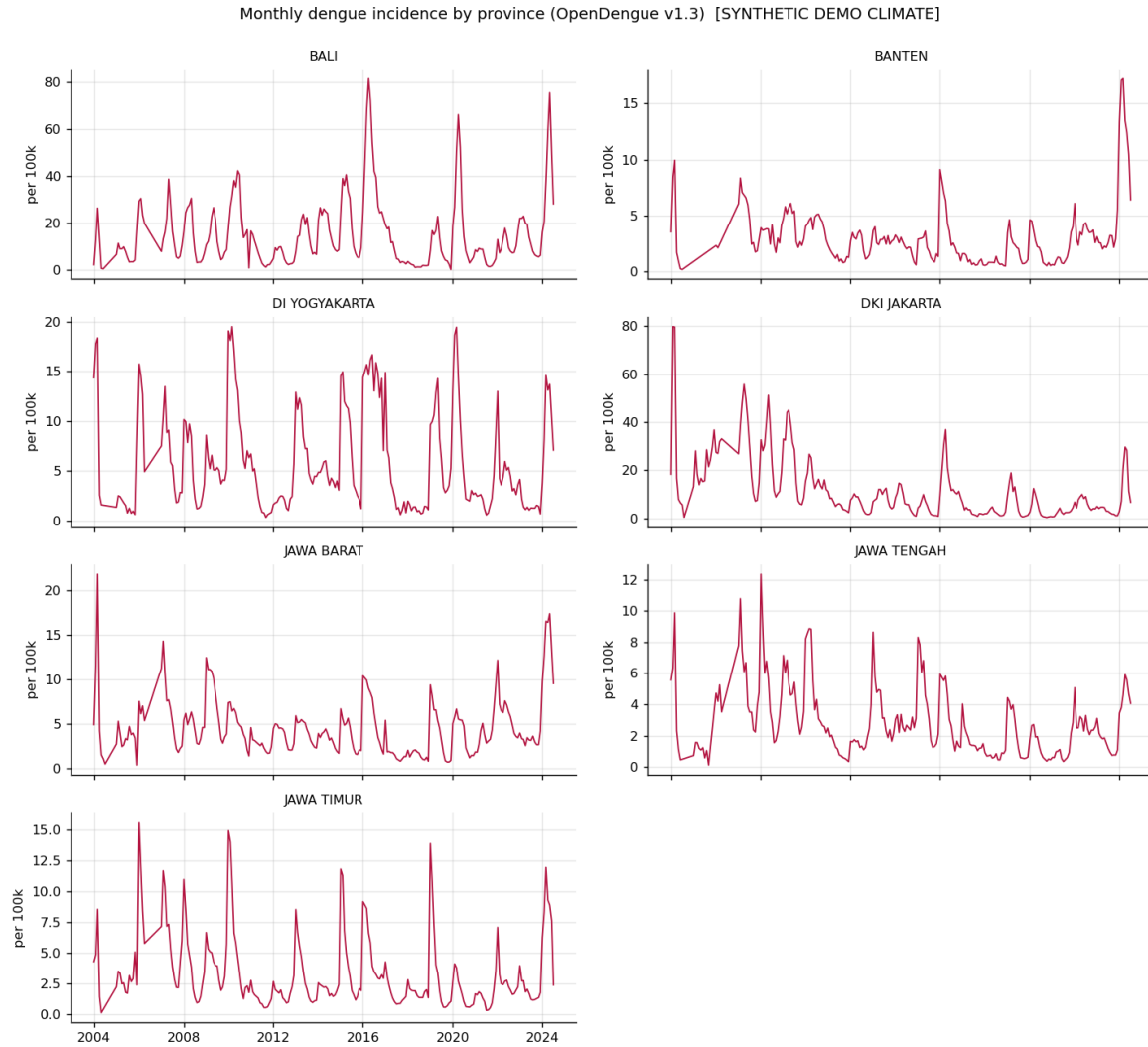


Figure 3: Monthly dengue incidence per 100,000 by province, 2004–2024.

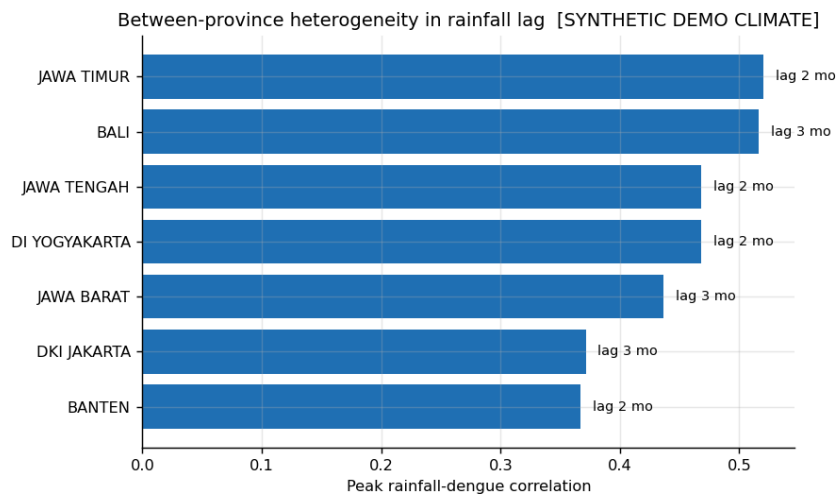


Figure 4: Between-province heterogeneity: peak rainfall–dengue correlation and the lag (months) at which it occurs.

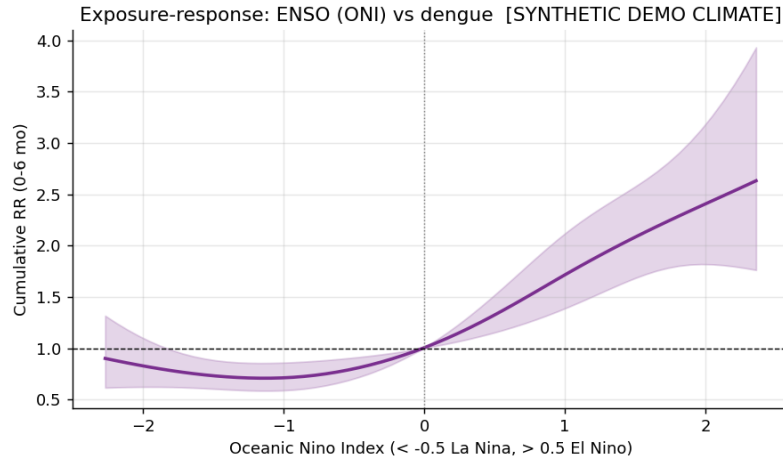


Figure 5: East Java cumulative exposure–response for ENSO (ONI), 0–6 month lag, relative to neutral conditions. Shaded band: 95% CI.

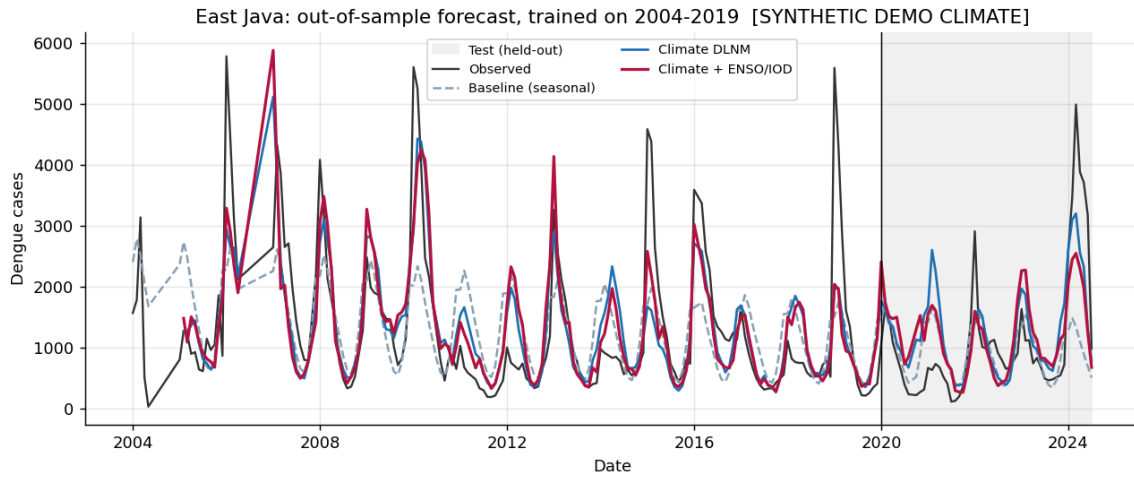


Figure 6: East Java out-of-sample forecast. The model is trained only on 2004–2019; the shaded region (2020–2024) is held out. The climate DLNM tracks the held-out epidemics more closely than the seasonal baseline.